

# Echelon form & Gaussian Elimination :

## Echelon Form & Row reduced echelon form

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### \* Motivation for Gaussian Elimination

→ Gaussian Elimination is a systematic method to solve system of Linear equation.

System of Linear equation

Transform →

Simpler, equivalent system that is easy to solve

→ Let's practice solving some problems based on Elimination Method

↑ School level  
Approach

Example: Find  $x$  and  $y$  for systems of Linear equation using Elimination method.

$$x + y = 5 \quad \text{--- ①}$$

$$x + 2y = 7 \quad \text{--- ②}$$

→  $e_1^n \text{ ①} - e_1^n \text{ ②}$

$$\begin{array}{r} x + y = 5 \\ - x + 2y = -7 \\ \hline -y = -2 \end{array}$$

$$y = 2$$

Now substitute  $y$  in eq<sup>n</sup> ①

$$x + y = 5$$

$$x = 5 - y = 5 - 2 = 3$$

$$(x, y) = (3, 2)$$

Before  $\begin{cases} x + y = 5 \\ x + 2y = 7 \end{cases} \Rightarrow \begin{cases} x + y = 5 \\ -y = -2 \end{cases}$  After

Triangular system

Example: Find  $(x, y, z)$  for systems of Linear equation using Elimination method.

$$2x + y + z = 5 \quad \text{--- ①}$$

$$4x - 6y = -2 \quad \text{--- ②}$$

$$-2x + 7y + 2z = 9 \quad \text{--- ③}$$

$$\begin{array}{rcl} \rightarrow \text{eq}^n \text{ ①} + \text{eq}^n \text{ ③} & & \\ 2x + y + z = 5 & & \\ -2x + 7y + 2z = 9 & & \\ \hline 8y + 3z = 14 & \text{--- ④} & \end{array}$$

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eq<sup>n</sup> ② - 2 times eq<sup>n</sup> ①

$$\begin{array}{rcl} 4x - 6y & = & -2 \\ 4x + 2y + 2z & = & 10 \\ \hline -8y - 2z & = & -12 \quad \text{--- ⑤} \end{array}$$

$e_5 \text{ ④} + e_5 \text{ ⑤}$

$$\begin{array}{r} -8y - 2z = -12 \\ 8y + 3z = 14 \\ \hline \boxed{z = 2} \quad \text{--- ⑥} \end{array}$$

Substitute  $z$  in ⑤

$$\begin{array}{r} -8y - 2 \times 2 = -12 \\ \boxed{y = 1} \end{array}$$

Substitute  $y$  and  $z$  in ①

$$\begin{array}{r} 2x + y + z = 5 \\ 2x + 1 + 2 = 5 \\ \boxed{x = 1} \end{array}$$

Before  $\begin{array}{l} 2x + y + z = 5 \\ 4x - 6y = -2 \\ -2x + 7y + 2z = 9 \end{array} \Rightarrow \begin{array}{l} 2x + y + z = 5 \\ -8y - 2z = -12 \\ z = 2 \end{array}$  After

Triangular System

Q. What if we need to solve a system of linear equation with 1000 equations and 1000 unknowns?

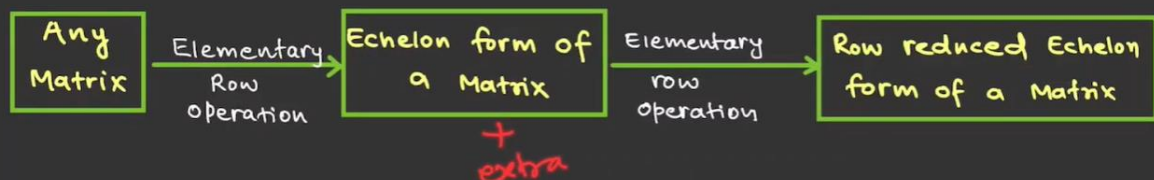
↳ Almost Impossible to solve using Elimination Method.

→ The elimination method we learned in school is a special, informal case of Gaussian elimination applied to small systems.

→ Whereas Gaussian Elimination is a systematic, matrix based algorithm that works for any size system.

→ In Gaussian Elimination, we work with matrix form

Idea → Use Elementary row operation to simplify the matrix until the solution becomes obvious.



Example:  $x + y = 3$   
 $3x - 2y = 4$

$\rightarrow Ax = b$

$$\begin{bmatrix} 1 & 1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

↑  
Coefficient Matrix

$$\left[ \begin{array}{cc|c} 1 & 1 & 3 \\ 3 & -2 & 4 \end{array} \right]$$

↑  
★ Augmented Matrix

\* Elementary Row operation:

→ Gaussian Elimination relies on elementary row operations which do not change the solution of systems of linear equation.

- ① swap 2 rows
- ② Multiply a row by a non-zero scalar
- ③ Add a multiple of one row to another row.

\* Echelon Form of a Matrix ← Similar to upper triangular matrix

→ A matrix is in echelon form if it satisfies all of the following conditions.

- ① All zero rows are below all non-zero rows.
- ② The first non-zero entry of each non-zero row is right to the first non-zero entry of previous row.

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- ③ All entries below each leading entry are 0

First non-zero entry of each non-zero row in echelon form is called Pivot (leading entry)

$$U = \begin{bmatrix} \bullet & * & * & * & * & * & * & * \\ 0 & \bullet & * & * & * & * & * & * \\ 0 & 0 & 0 & \bullet & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \bullet \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Example: Identify matrices in echelon form.

①  $\begin{bmatrix} 2 & * & * \\ 0 & 3 & * \\ 0 & 4 & * \end{bmatrix}$

②  $\begin{bmatrix} 5 & * & * & * \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix}$

③  $\begin{bmatrix} 5 & * & * & * \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

④  $\begin{bmatrix} 0 & 6 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}$

Example: Reduce the system to row echelon form.

$$x + 2y + z = 0$$

$$2x + 4y + 2z = 0$$

$$x + 2y + 3z = 0$$

→ Augmented Matrix =  $\left[ \begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 0 \\ 1 & 2 & 3 & 0 \end{array} \right]$

$$R_2 = R_2 - 2R_1$$

$$\left[ \begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 0 \end{array} \right] \xrightarrow{R_3 = R_3 - R_1} \left[ \begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right]$$

Gaussian Elimination

Forward Elimination

Back Substitution

$$R_2 \leftrightarrow R_3$$

$$\left[ \begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

\* Pivot, Pivot Column, Pivot Variable & Free Variable

→ In row-echelon form

① Pivot is the first non-zero entry in each non-zero row.

↳ It is a number

↳ Also called leading entry

② A pivot column is any column of a matrix that contains a Pivot

③ A pivot Variable (basic variable) is the variable corresponding to a pivot column.

④ A free Variable is a variable which is not a pivot variable.

→ System of equation corresponding to echelon form is

$$x + 2y + z = 0$$

$$2z = 0 \Rightarrow \boxed{z = 0}$$

→ substitute  $z$  in eq<sup>n</sup> ①

$$x = -2y$$

Identify free variable and assign constant parameter to that

→ free variable  $y = t$

$$x = -2t$$

$$\text{Solution } (x, y, z) = (-2t, t, 0)$$

$$\rightarrow \text{vector form of solution } \mathbf{x} = t \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

Example: solve the below system using Gaussian Elimination.

$$x_1 + x_2 - x_3 = 0$$

$$x_1 - x_2 + x_3 = 2$$

$$2x_1 - x_2 - x_3 = -3$$

$$\begin{aligned} \rightarrow & \left[ \begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 1 & -1 & 1 & 2 \\ 2 & -1 & -1 & -3 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - R_1} \left[ \begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -2 & 2 & 2 \\ 0 & -3 & 1 & -3 \end{array} \right] \\ & \downarrow R_3 \rightarrow R_3 - \frac{3}{2}R_2 \\ & \left[ \begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & -2 & 2 & 2 \\ 0 & 0 & -2 & -6 \end{array} \right] \end{aligned}$$

Here  $x_1, x_2, x_3 \rightarrow$  pivot variables

no free variables

→ Corresponding system of equations are

$$\begin{aligned}x_1 + x_2 - x_3 &= 0 \\ -2x_2 + 2x_3 &= 2 \\ -2x_3 &= -6\end{aligned}$$

from eq<sup>n</sup> ③  $x_3 = 3$

Substituting eq<sup>n</sup> ③ in ②

$$\begin{aligned}-2x_2 + 6 &= 2 \\ x_2 &= 2\end{aligned}$$

Substituting  $x_2$  and  $x_3$  in ①

$$\begin{aligned}x_1 + 2 - 3 &= 0 \\ x_1 &= 1\end{aligned}$$

Hence  $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$x_1, x_2, x_3$

### \* Row reduced echelon form ← Computing Inverse

→ A matrix is in row reduced echelon form if all of the following holds

- ① It is already in echelon form
- ② Each pivot is 1
- ③ all elements above pivot elements equal to 0

$$R = \begin{bmatrix} 1 & 0 & * & 0 & * & * & * & 0 \\ 0 & 1 & * & 0 & * & * & * & 0 \\ 0 & 0 & 0 & 1 & * & * & * & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

→ Simplest form of matrix that can be used to solve system of linear equations.